

Pawan Kumar Dangi

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FULL-STACK DEVELOPER & AI ENGINEER

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, MANIPUR B.TECH IN CSE(AI) | 2022 – PRESENT

TECHNICAL SKILLS

LANGUAGES & FRAMEWORKS

Python (Pandas, NumPy, Scikit-learn) • JavaScript • C++ • FastAPI • React.js • Node.js • Express.js • PostgreSQL • MySQL • Redis

TOOLS & COMPETENCIES

Docker • Streamlit • JWT • bcrypt • SendGrid • GitHub API • Render • RAG • LLM Agents • Full-Stack Dev • Risk Engine • API Design

PROJECTS

ADAPTIVE AUTHENTICATION SYSTEM FULL-STACK PRODUCTION APPLICATION

React + Vite + Tailwind CSS | FastAPI | PostgreSQL | Redis | SendGrid | Deployed on Render

- Built a production-grade authentication system with an intelligent risk engine that detects suspicious login patterns in real-time and triggers multi-factor verification.
- Engineered sophisticated risk scoring algorithm that analyzes 5+ behavioral signals: IP address tracking, geolocation analysis via ip-api.com, device fingerprinting (browser + OS), temporal anomalies, and failed login attempts.
- Implemented intelligent decision engine: Risk Score 0–29 grants immediate access, 30–99 requires OTP verification via email, 100+ blocks login automatically.
- Developed RESTful FastAPI backend featuring JWT token management (30-min expiry), bcrypt password hashing, Redis-based rate limiting (brute force protection), and PostgreSQL persistence with SQLAlchemy ORM.
- Created responsive React frontend with Vite bundling and Tailwind CSS styling: signup/login flows, OTP verification modal, secure dashboard with login history visualization, and risk score indicators.
- Implemented complete security pipeline: password hashing, JWT expiry, OTP auto-deletion (5-min window), device fingerprinting as MD5 hash, CORS validation, and HTTPS enforcement in production.
- Deployed full stack to Render with automated CI/CD: PostgreSQL managed database, Redis cache layer, SendGrid SMTP integration, environment-based configuration.
- Live Deployment: Frontend | API Docs (Swagger)

AUTONOMOUS INTELLIGENT ENGINEER (AIDE) AI-POWERED GITHUB ISSUE RESOLUTION AGENT

FastAPI | Streamlit | RAG + Vector Embeddings | LLM Agents | Docker Compose | GitHub Webhooks

- Architected an autonomous AI agent system that understands GitHub issues, analyzes codebases, generates fixes, and opens draft pull requests without human intervention.
- Implemented intelligent code retrieval using Retrieval-Augmented Generation (RAG): parses repository structure, builds vector embeddings for semantic search, and retrieves contextually relevant code files.
- Designed autonomous multi-step agent orchestration: (1) webhook receives issue, (2) clones repository, (3) parses codebase and builds retrieval index, (4) identifies root cause through RAG, (5) generates repair plan, (6) applies code modifications, (7) validates syntax, (8) runs tests if present, (9) creates feature branch and draft PR.
- Built dual-interface system: automated GitHub webhook handler (FastAPI on port 8000) for production workflows and manual Streamlit UI (port 18601) for interactive code fixing without webhooks.
- Integrated multi-LLM provider support: Ollama (local), OpenAI, Gemini, Groq, Anthropic, and Grok with flexible provider switching via environment configuration.
- Containerized entire application with Docker Compose: isolated FastAPI backend, Streamlit frontend, and optional Ollama service for reproducible local development and deployment.
- Implemented GitHub integration: repository cloning, issue parsing, branch creation, commit signing, and draft PR generation with automated comments linking back to original issue.
- Repository: github.com/pwnFirstGit